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SUBJECT CODE NO: - RNP-8048
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - IV
Examination May/June-2026
AC Machines

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION – A**20****Q.1 Solve any 10 from the following questions.**

- 1) The V-curves of synchronous motor represent relation between:

A) Speed & torque	B) Voltage & current
C) Armature current vs field current	D) Power & speed
- 2) In an alternator, armature winding is placed on:

A) Rotor	B) Stator	C) Both	D) Shaft
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- 3) Cogging in induction motors occurs due to:

A) High voltage	B) Low current
C) Magnetic locking	D) Overheating
- 4) In a Synchronous motor, the rotor runs at:

A) Less than synchronous speed	B) Greater than synchronous speed
C) Synchronous speed	D) Zero speed
- 5) Hunting in synchronous motors refers to:

A) Noise	B) Heating
C) Oscillation of rotor	D) Overloading
- 6) The slip of an induction motor at standstill is:

A) 0	B) 0.5	C) 1	D) -1
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- 7) Maximum torque in an induction motor occurs when:

A) Slip is zero	B) Rotor resistance equals reactance
C) Slip is unity	D) Voltage is zero
- 8) Which starter is used for 3-phase induction motors?

A) DOL starter	B) Star-delta starter
C) Auto-transformer starter	D) All of these

- 9) The speed of an induction motor can be controlled by:
 - A) Rotor resistance
 - B) Stator voltage
 - C) Frequency control
 - D) All of these
- 10) The equivalent circuit of an induction motor resembles:
 - A) Transformer
 - B) DC motor
 - C) Generator
 - D) Battery
- 11) The power factor of a synchronous motor can be controlled by:
 - A) Load
 - B) Speed
 - C) Excitation
 - D) Voltage
- 12) Which motor is self-starting?
 - A) Synchronous motor
 - B) Induction motor
 - C) Alternator
 - D) Generator
- 13) The torque in an induction motor is proportional to:
 - A) Slip
 - B) Rotor current
 - C) Power factor
 - D) Both A and B
- 14) Single phase induction motor is not self-starting because of:
 - A) High resistance
 - B) Low voltage
 - C) Double field revolving theory
 - D) High current

SECTION – B

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| Q.2 | a) Explain the torque-slip characteristics of an induction motor. | 05 |
| | b) Explain the voltage regulation of an alternator using the EMF method with a neat diagram. | 05 |
| Q.3 | a) Explain V-curves and inverted V-curves of a synchronous motor. | 05 |
| | b) Describe the applications of three-phase synchronous motors. | 05 |
| Q.4 | a) Discuss the synchronization of alternators and state the conditions required. | 05 |
| | b) Derive the torque equation of a three-phase induction motor | 05 |
| Q.5 | a) Describe the no-load and blocked rotor tests of an induction motor. | 05 |
| | b) Discuss the methods of speed control of induction motors. | 05 |
| Q.6 | a) Explain the construction and working of a single-phase induction motor using Double field revolving theory | 05 |
| | b) Describe the different types of single-phase induction motors and their applications. | 05 |
| Q.7 | a) Explain the alternator on load with resistive, inductive, and capacitive loads using phasor diagrams. | 05 |
| | b) Explain the equivalent circuit of a three-phase induction motor. | 05 |